Truly Subquadratic 3SUM over Every Bounded-Exponent Abelian Group

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For every fixed integer $M \geq 2$, we give a randomized truly subquadratic algorithm for three-list zero-sum matching over every compactly represented finite abelian group of exponent dividing M and polynomial order. The exponent saving depends on M and is independent of the polynomial order bound. The algorithm has a worst-case time cap, returns a verified witness, and succeeds with probability at least $3/4$ on every fixed positive input. It extends the known exponent-dividing-p^2 scope to all higher prime powers and mixed-prime products. The main step combines simultaneous profiles of original annihilator-coset sizes with an extremal divisor certificate. The certificate controls both class branching and false positives of every possible sum order, enabling one elementary-middle support product per profile. We use the existing selected-entry thin-matrix algorithm of Alman and Vassilevska Williams. Through Majdoddin's finite-representation reduction, the result gives a truly subquadratic algorithm for every fixed abelian-group-representable finite coordinate relation.

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1 The result and its mechanism

Three-list 3SUM asks whether $x_1 + x_2 + x_3 = 0$ for some x_i in three given lists. The ambient additive group affects the available hashing: a uniform homomorphism can preserve a vector of order two much more often than one of large order. This makes an order-independent collision estimate unsuitable for groups with several torsion layers. We resolve the bounded-exponent question by adapting the hash to all order layers simultaneously.

The input group is given as $G_0 = \prod_{j=1}^{r} \mathbb{Z}_{m_j}$, with each $m_j \mid M$, and list elements are residue vectors. Trivial factors are discarded. For every fixed $c > 0$ we allow $|G_0| \le n^c$. Costs include group arithmetic, construction of every auxiliary list and matrix, all queries, random-bit generation, and recovery of original list indices. A charged word has $O_{M,c}(\log n)$ bits; bit simulation contributes a polynomial logarithmic factor.

Theorem 1. *Fix an integer $M \ge 2$, and put*

$$C = \lceil \log_2 M \rceil, \qquad \eta = \min\{1/18, 1/(8C)\}, \qquad \gamma = 63/1000, \qquad \delta_M = \eta\gamma/2.$$

There is a uniform classical randomized algorithm which, for every fixed $c > 0$, solves three-list 3SUM over every such G_0 of order at most n^c and lists of size at most n in worst-case time $O_{M,c}(n^{2-\delta_M} \operatorname{polylog} n)$. It always rejects negative inputs, returns only verified witnesses, and returns a witness on each fixed positive input with probability at least $3/4$.

Majdoddin's Theorem 19.17 and Corollary 19.25 give deterministic algorithms for the entire exponent-dividing-p^2 class, for each fixed prime p; Open Problem 19.26 asks for the general bounded-exponent extension [Maj26]. Theorem 1 supplies that extension with randomness and a smaller explicit saving. For example, its scope includes arbitrary products of $\mathbb{Z}_8$, and products with different prime divisors, with ranks growing logarithmically with n.

The support-product construction and its selected-entry matrix engine come from prior work. Alman and Vassilevska Williams supply the thin-product algorithm [AVW26]; Majdoddin's Lemmas 19.6–19.7 organize homomorphic matching as support products and probes [Maj26]. The contribution here is the simultaneous choice of quotient and row counts. We profile original coset occupancies for every divisor of M. A least feasible row count gives lower bounds for all divisors and an upper bound witnessed by one divisor. The former controls every false-positive order, while the latter controls the number of hash classes. The two bounds make the inherited support construction work for the full bounded-exponent class.

The resulting tractability principle also covers every fixed finite coordinate relation represented by an equation in an abelian group (Corollary 8). Its finite-quotient and translation arguments are Majdoddin's Lemmas 20.3(d) and 20.4(a); Remark 20.6 explicitly anticipates this consequence of the general group algorithm [Maj26]. Recent work of Chan, Itakura, Ma and Yu gives another truly subquadratic approach to integer 3SUM through subset convolution [CIMY26]. Our algorithm uses the thin-product route and its group-specific balancing argument.

Research methodology. OpenAI Codex materially contributed to the divisor-profile proof, literature inspection, exact finite verification code, and manuscript drafting. A fresh model context checked the proof and implemented separate finite checks. These roles affect the mathematical and algorithmic sections throughout; the final AI methodology statement gives further detail.

2 Normalization and original-coset profiles

Deduplicate each list, retaining an original index for each value. Embed $\mathbb{Z}_m$ into $\mathbb{Z}_M$ by $x \mapsto (M/m)x$. This is an injective homomorphism, so the embedded lists have exactly the same zero-sum witnesses. We henceforth work in $G = \mathbb{Z}_M^r$. Since $r \le c \log_2 n$, the enlarged order is at most $n^{c \log_2 M}$ and each evaluation or comparison of a residue vector has polynomial logarithmic cost. The trivial group and empty lists are handled directly.

If one list has size at most $\sqrt{n}$, enumerate its pairs with another list and use a sorted lookup in the third. The cost is $O_{M,c}(n^{3/2} \operatorname{polylog} n)$. Assume below that all three original normalized lists S_i have size greater

than $\sqrt{n}$. We also assume $n \geq 16$ and $R \geq M$ for the parameter R defined in Section 3. For fixed M the excluded values of n are bounded, so exhaustive search on them is absorbed into the claimed bound.

Let $\mathcal{D} = \{d \geq 2 : d \mid M\}$ and $L = |\mathcal{D}|$. For $d \in \mathcal{D}$ write

$$H_d = G[d] = \{x : dx = 0\} = (M/d)G.$$

Reduction of coordinates modulo M/d represents the quotient by H_d. For each $x \in S_i$, let $c_{i,d}(x)$ be the number of elements of the *original* S_i in $x + H_d$, and let $s_{i,d}(x)$ be the largest power of two not exceeding $c_{i,d}(x)$. Partition S_i by the full vector $(s_{i,d}(x))_{d\in\mathcal{D}}$. Sorting computes all original coset counts and these profiles in $O_{M,c}(n \operatorname{polylog} n)$ time. There are at most $(1 + \lfloor \log_2 n \rfloor)^L$ profiles per list.

Process every triple of nonempty profiles, obtaining a subinstance (U_1, U_2, U_3). Every original triple belongs to exactly one such subinstance. Within it, $s_{i,d}$ is constant; define $s_d = \min_i s_{i,d}$. The following two different uses of original occupancy are the reason the profile restriction is safe.

Lemma 2. *For a fixed subinstance and each $d \in \mathcal{D}$:*

(1) *the number T_d of triples in $U_1 \times U_2 \times U_3$ with sum in H_d is less than $2s_d n^2$;*
(2) *the number of original H_d-cosets represented in U_i is at most $n/s_{i,d} \leq n/s_d$.*

Moreover $s_M > \sqrt{n}/2 \geq n^{1/4}$.

Proof. Choose a list attaining s_d. Fixing the two other elements determines one coset for its element. Any represented coset has fewer than $2s_d$ elements in its original list, and restriction cannot increase this count. This proves the first assertion. For the second, each represented original coset contains at least $s_{i,d}$ original elements; these cosets are disjoint. Finally $H_M = G$, so each $s_{i,M}$ is the dyadic rounding of $|S_i| > \sqrt{n}$. The last inequality follows from $n \geq 16$. □

A restricted coset may contain only one element, even when its original occupancy is large. Lemma 2 uses an upper bound on restricted occupancy and a lower bound on original occupancy; it never needs a lower bound on restricted occupancy.

3 The divisor certificate and the hash

For a fixed profile subinstance set

$$h = \lfloor \eta \log_2 n \rfloor, \quad R = 2^h, \quad \alpha = \tfrac{1}{2} - \tfrac{\gamma}{4} = \tfrac{1937}{4000}, \quad \sigma = \lfloor R^\alpha \rfloor. \tag{1}$$

Thus $n^\eta/2 \leq R \leq n^\eta$ and $R^\alpha/2 \leq \sigma \leq R^\alpha \leq \sqrt{R}$. Choose the least nonnegative integer ℓ satisfying

$$d^\ell \sigma \geq s_d R \qquad (d \in \mathcal{D}). \tag{2}$$

Because $s_M \geq n^{1/4}$ and $R/\sigma \geq 1$, $\ell \geq 1$. Minimality supplies $e \in \mathcal{D}$ for which the inequality fails at $\ell - 1$, and hence

$$e^\ell < e s_e R/\sigma. \tag{3}$$

This is the extremal divisor certificate: all divisors have the lower bound (2), while one divisor has the complementary upper bound (3). It is computed from the profile values in $O_M(\log n)$ integer-power comparisons, and $\ell = O_M(\log n)$.

Choose a prime $p \mid e$, and put

$$q = \left\lfloor \log_p R \right\rfloor, \qquad D = p^q, \qquad k = \ell - q.$$

Our large-n assumptions imply $q \geq 1$ and $R/M < D \leq R$. Also

$$\ell \geq \tfrac{1}{4} \log_M n, \qquad q \leq \eta \log_2 n \leq \tfrac{1}{8} \log_M n,$$

so $k \geq 0$. All parameters are exact integers. For example, rational-power rounding in (1) is implemented by integer-power comparisons on $O_M(\log n)$-bit numbers.

Take independent uniform rows $u_1, \dots, u_\ell \in \mathbb{Z}_M^r$. Define two homomorphisms by

$$\pi(x) = \big(x + H_e, (\langle u_j, x\rangle \bmod M)_{j\le k}, (\langle u_j, x\rangle \bmod (M/p))_{k<j\le \ell}\big), \tag{4}$$

$$K(x) = \big(x + H_e, (\langle u_j, x\rangle \bmod M)_{j\le \ell}\big). \tag{5}$$

The value of K determines π. The quotient component is stored as a residue vector; its entire group is never enumerated. Each evaluation has $O_{M,c}(\log^2 n)$ cost. If $x \in \ker \pi$, the quotient component and the first k coordinates of $K(x)$ vanish, and each remaining coordinate is a multiple of M/p. Consequently

$$K(\ker \pi) \subseteq B = \{0\} \times \{0\}^k \times ((M/p)\mathbb{Z}_M)^q \cong \mathbb{F}_p^q, \qquad |B| = D. \tag{6}$$

Using this whole explicit subgroup is sufficient, even if the actual image is smaller. Split its coordinates as $B = B_1 \times B_2$ with dimensions $\lceil q/2\rceil$ and $\lfloor q/2\rfloor$.

Lemma 3. *For every choice of rows, each U_i has fewer than $M^2 n/\sigma$ nonempty π-classes. Cutting each such class into blocks of size at most σ gives at most $(M^2+1)n/\sigma$ blocks.*

Proof. On one H_e-coset a full row has at most e values. A row reduced modulo M/p has at most e/p values, since $p \mid e$. Thus at most

$$e^k (e/p)^q = e^\ell/D < e s_e R/(\sigma D) < M^2 s_e/\sigma$$

distinct π-labels occur in that coset. Multiply by at most n/s_e represented original cosets, using Lemma 2. The number of blocks is at most the number of classes plus $|U_i|/\sigma$. □

4 Controlling every sum order

Lemma 4. *If $v \in G$ has additive order d, a uniform row maps v uniformly onto the subgroup of $\mathbb{Z}_M$ of order d. For nonzero $v \in H_e$,*

$$\mathbb{P}[K(v) = 0] = d^{-\ell}, \qquad \mathbb{P}[\pi(v) = 0] = \gcd(d,p)^q d^{-\ell} \le D d^{-\ell}. \tag{7}$$

Vectors outside H_e have survival probability zero under both maps.

Proof. The map $u \mapsto \langle u, v\rangle$ is a homomorphism. Its image is the ideal generated by the coordinates of v in $\mathbb{Z}_M$, of size d; all fibers have equal size. Reduction modulo M/p takes that image to one of size $d/\gcd(d,p)$. Independence of the rows gives (7). The common quotient component excludes vectors outside H_e. □

Let Z_0 count true zero-sum triples in the subinstance, let Z_π count triples whose sum is in $\ker \pi$, and let F count triples of nonzero sum in $\ker K$. Every nonzero element of H_e has order $d > 1$ dividing e. The number of triples of exact sum-order d is at most T_d, hence less than $2 s_d n^2$. Lemma 4 and (2) give

$$\mathbb{E}F \le \sum_{\substack{d\mid e\\ d>1}} 2 s_d n^2 d^{-\ell} \le 2L\sigma n^2/R, \tag{8}$$

$$\mathbb{E}(Z_\pi - Z_0) \le D\,\mathbb{E}F \le 2L\sigma n^2. \tag{9}$$

Markov's inequality and a union bound imply that with probability at least $5/6$ both

$$F \le 24 L\sigma n^2/R, \qquad Z_\pi - Z_0 \le 24 L\sigma n^2 \tag{10}$$

hold. Dependence between these two random counts is harmless. Since lists are sets, $Z_0 \le n^2$. We call (10) the good event. The counts in its definition are used only in the analysis. The algorithm observes the listing and scanning caps below instead of evaluating these counts.

5 Support products, probes, and recovery

We spell out the support construction to account for all queries and witness scans. The construction follows the framework of Majdoddin's Lemmas 19.6–19.7 [Maj26]. Form the blocks in lists 2 and 3 given by Lemma 3, and choose a representative z_b in every block. A probe is $Q = (x_1; b_2, b_3)$ with $x_1 \in U_1$ and

$$\pi(x_1) + \pi(z_{b_2}) + \pi(z_{b_3}) = 0.$$

For every pair (x_1, b_2), look up the compatible label of list 3 and enumerate its blocks. Sorted label indices implement this in polynomial logarithmic time per pair and output.

Lemma 5. *Let C_2 be the number of nonempty π-classes in U_2. The set $\mathcal{P}$ of probes satisfies*

$$|\mathcal{P}| \le nC_2 + 2n^2/\sigma + Z_\pi/\sigma^2. \tag{11}$$

On the good event, $|\mathcal{P}| \le An^2/\sigma$ for $A = M^2 + 3 + 24L$.

Proof. Fix x_1. For a compatible pair of nonempty classes of sizes a, b, the block-pair count is at most $(1 + a/\sigma)(1 + b/\sigma)$. Summing the constant term over all x_1 and labels gives at most nC_2. The two linear terms each sum to at most n^2/σ: translation of labels is injective, so each class of either list appears at most once for a fixed x_1. The bilinear term sums to Z_π/σ^2. On the good event, $Z_\pi \le n^2 + 24L\sigma n^2$; use Lemma 3 and $\sigma \ge 1$. □

During listing, abort this subinstance if the next probe would exceed $\lceil An^2/\sigma \rceil$. This cap is observable. The outer enumeration costs $O_M(n^2/\sigma)$, plus polynomial logarithmic work per stored probe. An aborted subinstance contributes no witness and processing continues.

For a nonaborted subinstance put

$$b_0 = M^2 + 1, \qquad g_0 = M(A + b_0 + 3), \qquad g' = \lceil g_0 n/\sigma \rceil, \qquad N = g' p^{\lceil q/2 \rceil}.$$

Left vertices are pairs (b_2, μ_1) with $\mu_1 \in B_1$; right vertices are (b_3, μ_2) with $\mu_2 \in B_2$. Pad both sides to size N, and use B as a middle set. Lemma 3 ensures sufficient room. Construct 0/1 matrices $X \in \mathbb{Z}^{N \times D}$ and $Y \in \mathbb{Z}^{D \times N}$ by

$$X[(b_2, \mu_1), \lambda] = 1 \iff \exists x_2 \in b_2 : \; K(x_2) - K(z_{b_2}) + (\mu_1, 0) = \lambda, \tag{12}$$

$$Y[\lambda, (b_3, \mu_2)] = 1 \iff \exists x_3 \in b_3 : \; -(K(x_3) - K(z_{b_3})) - (0, \mu_2) = \lambda. \tag{13}$$

All differences in these definitions lie in B, because a block lies within one π-class. For a probe Q the element

$$w_Q = K(x_1) + K(z_{b_2}) + K(z_{b_3}) \in B$$

has a unique decomposition (μ_1, μ_2). Query the product at $((b_2, \mu_1), (b_3, \mu_2))$. Deduplicate these matrix positions while retaining the corresponding probe lists.

Lemma 6. *The product entry for a probe is positive exactly when some pair $x_2 \in b_2$, $x_3 \in b_3$ satisfies $K(x_1 + x_2 + x_3) = 0$. Every true witness has a positive probe, and at most F positive probes contain no true witness.*

Proof. A positive integer product entry means that the two support equations (12)–(13) hold for one common λ. Equating them gives

$$K(x_2) + K(x_3) - K(z_{b_2}) - K(z_{b_3}) + (\mu_1, \mu_2) = 0.$$

Substituting w_Q yields precisely $K(x_1 + x_2 + x_3) = 0$. Each step is reversible. A true witness has zero π-sum and zero K-sum, and thus gives a positive probe. A positive probe with no true witness contains a nonzero-sum K-zero triple. Distinct probes contain disjoint triples, since a triple determines x_1 and both blocks. Choosing one such triple per failed probe bounds their number by F, even if probes share a matrix position. □

Scan a positive probe by enumerating $x_2 \in b_2$ and looking up $-x_1 - x_2$ in the sorted block b_3. This uses original residue vectors, costs $O_{M,c}(\sigma \operatorname{polylog} n)$, and returns original indices when successful. Scan at most $\lceil 24L\sigma n^2/R \rceil + 1$ positive probes, stopping immediately on a verified witness. If all these scans fail, abort the subinstance. On the good event, Lemma 6 ensures that a subinstance containing a witness finds one within this cap. A final direct sum check verifies every returned witness.

6 The legal matrix call and the total bound

We use the following existing algorithmic theorem, including its stated integer-input and selected-entry interface.

Lemma 7 (Alman–Vassilevska Williams, Corollary 26 [AVW26]). *Let $N \geq D^{18}$, and let $X \in \mathbb{Z}^{N\times D}$ and $Y \in \mathbb{Z}^{D\times N}$ have entries of absolute value $N^{O(1)}$. They can be preprocessed deterministically in $O(N^2/D^{0.063})$ time and space, after which an entry of XY can be queried in $O(D^{0.437})$ time. In particular, at most $N^2/\sqrt{D}$ prescribed entries can be computed in $O(N^2/D^{0.063})$ time.*

Here the entries of X, Y are $0/1$. Since $\sigma \leq \sqrt{R}$ and $D > R/M$,

$$N \geq g_0 n\sqrt{D}/\sigma \geq n.$$

Consequently $D \leq R \leq n^\eta \leq N^{1/18}$. Also $D^{1+\gamma} \leq N$, since $\eta \leq 1/18$ and $\gamma < 1/2$. The listing cap is at most $(A+1)n^2/\sigma$, and

$$\frac{N^2}{\sqrt{D}} \geq \frac{g_0^2 n^2\sqrt{D}}{\sigma^2} \geq \frac{g_0^2}{\sqrt{M}}\frac{n^2}{\sigma} \geq (A+1)\frac{n^2}{\sigma}. \tag{14}$$

Thus one call to Lemma 7 computes all required positions.

For the cost, $p^{\lceil q/2\rceil} \leq \sqrt{pD}$ and $g' \leq (g_0+1)n/\sigma$. Since D and R differ by at most a fixed factor, the matrix call costs

$$O_M(N^2/D^\gamma) = O_M(n^2R^{1-\gamma}/\sigma^2) = O_M(n^2R^{-\gamma/2}). \tag{15}$$

Initializing the two dense input matrices takes $O(ND)$ time, covered by this bound because $D^{1+\gamma} \leq N$. Set their nonzero entries by enumerating each list element and all possible μ_i. This costs $O_{M,c}(n\sqrt{pD}\operatorname{polylog} n)$, also covered by (15). Sorting the capped probe positions and retaining their lists costs $O_{M,c}(n^2\sigma^{-1}\operatorname{polylog} n)$. This fits because $\alpha \geq \gamma/2$.

The capped scans, including the possible final successful scan, cost

$$O_{M,c}\big((\sigma^2n^2/R + \sigma)\operatorname{polylog} n\big) = O_{M,c}(n^2R^{-\gamma/2}\operatorname{polylog} n). \tag{16}$$

The equality in the dominant exponent follows from $2\alpha - 1 = -\gamma/2$. Row evaluation, class sorting, label arithmetic, parameter computation and recovery use $O_{M,c}(n\operatorname{polylog} n)$ additional time. Matrix dimensions are polynomial in n, so word sizes and their bit simulation add polynomial logarithmic overhead only.

For completeness, uniform coefficients in $\mathbb{Z}_M$ are generated from C fair bits by rejecting values at least M. An accepted coefficient needs at most two attempts in expectation. For each profile subinstance, cap the total number of attempts for its $r\ell$ coefficients at $24r\ell$, aborting the subinstance if the cap is exceeded. This cap costs $O_M(r\ell)$ time on every execution. Coupling with unlimited independent uniform generation, Markov's inequality bounds the probability of such an abort by $1/12$.

Every subinstance therefore has cost $O_{M,c}(n^2R^{-\gamma/2}\operatorname{polylog} n)$ on every random path, whether or not its listing or scans abort. There are at most $(1 + \lfloor \log_2 n \rfloor)^{3L}$ subinstances. As M is fixed and $R \geq n^\eta/2$, their total cost is the bound in Theorem 1, with saving $\delta_M = \eta\gamma/2$ independent of c.

To verify correctness, fix a witness of a positive input before any randomness. Its profile subinstance is deterministic. Its ideal uniform rows satisfy the two inequalities in (10) except with probability at most $1/6$. The coefficient cap has probability at most $1/12$ of interfering. On the complementary event the listing cap holds, and a positive probe containing a true witness is reached within the scan cap. Thus this subinstance finds a witness with probability at least $3/4$. If the algorithm stops earlier it has already found a verified witness. One may regard independent random tapes for all subinstances as sampled in advance, so stopping

early does not require a union bound over them. Negative inputs always return absence because every success requires an original zero-sum triple. This proves Theorem 1.

7 Finite coordinate relations

Let Σ be a fixed finite alphabet. A relation $P \subseteq \Sigma^3$ is abelian-group-representable if there are maps $\phi_i : \Sigma \to A$ and $t \in A$ into an abelian group with

$$(a, b, c) \in P \quad \Longleftrightarrow \quad \phi_1(a) + \phi_2(b) + \phi_3(c) = t.$$

Given three lists in Σ^d, the coordinatewise problem asks for one triple whose coordinate triples all belong to P.

Corollary 8. *For each fixed abelian-group-representable finite relation P, there is $\delta_P > 0$ such that, for every fixed $c > 0$, its coordinatewise problem on lists of size at most n and $d \le c \log n$ has a randomized witness algorithm of worst-case time $O_{P,c}(n^{2-\delta_P} \operatorname{polylog} n)$, with the one-sided guarantee of Theorem 1.*

Proof. We give the finite-quotient argument of Majdoddin's Lemma 20.3(d), followed by the translation of Lemma 20.4(a) [Maj26]. The empty relation is immediate. Otherwise fix $s^0 \in P$ and use the basis $e_{i,a}$ of $\mathbb{Z}^{3|\Sigma|}$. Put

$$e(s) = e_{1,s_1} + e_{2,s_2} + e_{3,s_3}, \qquad L_P = \operatorname{span}_{\mathbb{Z}}\{e(s) - e(s^0) : s \in P\}.$$

Any representation induces a homomorphism vanishing on L_P, so for $s \notin P$ the vector $e(s) - e(s^0)$ is outside L_P. In a Smith normal form basis, write $L_P = \bigoplus_{i \le \rho} d_i \mathbb{Z} e'_i$. There are only finitely many forbidden vectors. Choose an integer $m \ge 2$ divisible by every d_i and larger than the absolute values of all coordinates of these vectors. A forbidden vector y remains outside $L_P + m\mathbb{Z}^{3|\Sigma|}$: either some $i \le \rho$ has $d_i \nmid y_i$, or some $i > \rho$ has nonzero y_i, which cannot be divisible by m. The finite group

$$A_P = \mathbb{Z}^{3|\Sigma|} / (L_P + m\mathbb{Z}^{3|\Sigma|})$$

therefore represents exactly P, using $\phi_i(a) = [e_{i,a}]$ and $t = [e(s^0)]$. This group and its exponent depend only on P.

For a string $x \in \Sigma^d$ define $\Phi_i(x) = (\phi_i(x[j]))_{j=1}^d$. Translate the third list by $-(t, \dots, t)$ and leave the other two image lists unshifted. A resulting zero sum in A_P^d is equivalent, coordinate by coordinate, to a valid original triple. The construction costs $O_P(nd)$; retain a preimage index for every image value. These indices recover a valid original witness even when image strings coincide. Since $|A_P^d| \le n^{O_P(c)}$ and its exponent is fixed, apply Theorem 1 with $M = \max\{2, \exp(A_P)\}$ and $\delta_P = \delta_M$. The exponent saving is independent of c. □

This supplies the sufficient tractability direction for every representable relation. The argument leaves the complexity of nonrepresentable relations as a separate question. The prior finite-quotient reduction accounts for the breadth of the consequence; the group algorithm supplies the extension that it requires.

8 Verification and further questions

The proof gives an explicit, bounded-time reduction to the prior thin-product algorithm. Its finite checks supplement the algebra: the separate checker tested 778 full-row and 1,382 reduced-row distributions, 4,622 profile subinstances, and 144 literal support instances containing 45,932 matrix entries and 13,244 probes. The tests included mixed moduli, fragmented profiles and nonsurjective images. These are regression tests; the bounds for all inputs follow from the lemmas above. The thin-product algorithm is used as a cited theorem. A faster saving as M grows and a deterministic replacement for the row-selection step remain useful directions.

References

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AI methodology statement

OpenAI Codex generated and materially influenced the proof development, literature comparisons, verification programs and English exposition in this paper. Codex used primary-source text and web metadata for the cited results. A separate fresh model context reconstructed the proof and wrote a second exact checker; finite tests were executed locally. The manuscript was drafted and compiled with Codex assistance. Proof verification consisted of model-based mathematical reconstruction and local exact finite checks. The sole listed author is responsible for the paper's content.